Creating Visualisations for the Blog

Below I am setting up colour palettes to use in the visualisations to get consistent mappings throughout the plots.

```
## importing packages
import pandas as pd
import re
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from IPython.display import HTML

## creating the mapping for countries to be used in both the book and series
plots
palette16 = dict(Chinese = "pink", Dutch = "orchid", English = "thistle", French
= "mediumslateblue", German ="cornflowerblue",
             Italian = "lightsteelblue", Japanese = "lightskyblue", Norwegian
="mediumaquamarine", Polish = "lightgreen", Russian = "lightgoldenrodyellow",
        Swedish = "moccasin", Portuguese = "sandybrown", Spanish = "darksalmon",
Czech = "lightcoral",Yiddish = "silver",
           Gujarati = "darkgrey")

## this below palette I will be using for genres in the book genre plot
palette15 = ( "pink", "orchid",  "thistle",  "mediumslateblue", "cornflowerblue",
          "lightsteelblue", "lightskyblue", "mediumaquamarine",  "lightgreen",
"lightgoldenrodyellow",
            "moccasin",  "sandybrown",  "darksalmon",  "lightcoral", "silver")
```

```
/opt/anaconda3/lib/python3.9/site-packages/pandas/core/computation/
expressions.py:21: UserWarning: Pandas requires version '2.8.4' or newer of
'numexpr' (version '2.8.3' currently installed).
  from pandas.core.computation.check import NUMEXPR_INSTALLED
/opt/anaconda3/lib/python3.9/site-packages/pandas/core/arrays/masked.py:60:
UserWarning: Pandas requires version '1.3.6' or newer of 'bottleneck' (version
'1.3.5' currently installed).
  from pandas.core import (
/opt/anaconda3/lib/python3.9/site-packages/scipy/__init__.py:155: UserWarning:
A NumPy version >=1.18.5 and <1.25.0 is required for this version of SciPy
(detected version 1.26.4
  warnings.warn(f"A NumPy version >={np_minversion} and <{np_maxversion}"
```

This section will focuse on visualisations using the individual book data.

```
## loading in the book dataframe and checking the formating is correct
df_book = pd.read_csv('../data/top_books.csv')
df_book = df_book.drop(df_book.columns[0], axis =1)
df_book
```

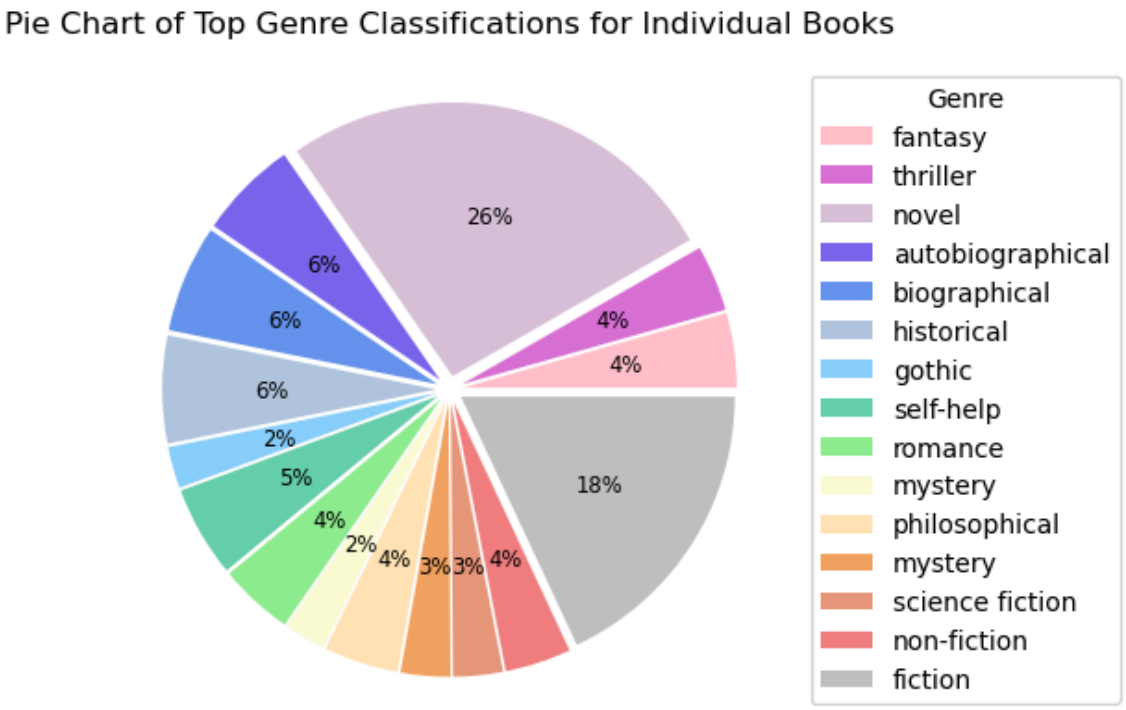
	Book	Author	Original_Language	First_Published	Approx_Sales	Genre_1	Genre_2	Genre_3	Genre_4	Genre_5	Genre_6	All_Genre	Age_Category
0	A Tale of Two Cities	Charles Dickens	English	1859	200000000	Historical fiction	NaN	NaN	NaN	NaN	NaN	Historical fiction	Adult
1	The Little Prince (Le Petit Prince)	Antoine de Saint-Exupéry	French	1943	200000000	Fantasy	children's fiction	NaN	NaN	NaN	NaN	Fantasy, children's fiction	Children
2	The Alchemist (O Alquimista)	Paulo Coelho	Portuguese	1988	150000000	Fantasy	NaN	NaN	NaN	NaN	NaN	Fantasy	Adult
3	Harry Potter and the Philosopher's Stone	J. K. Rowling	English	1997	120000000	Fantasy	children's fiction	NaN	NaN	NaN	NaN	Fantasy, children's fiction	Children
4	And Then There Were None	Agatha Christie	English	1939	100000000	Mystery	NaN	NaN	NaN	NaN	NaN	Mystery	Adult
...	...	...	...	...	...	...	...	...	...	...	...	...	...
161	Bridget Jones's Diary	Helen Fielding	English	1996	10000000	NaN	NaN	NaN	NaN	NaN	NaN	novel	Adult

	Book	Author	Original_Language	First_Published	Approx_Sales	Genre_1	Genre_2	Genre_3	Genre_4	Genre_5	Genre_6	All_Genre	Age_Category
162	It Ends with Us	Colleen Hoover	English	2016	10000000	Romance	Fiction	NaN	NaN	NaN	NaN	Romance, Fiction	Adult
163	The Lion, the Witch and the Wardrobe	C.S. Lewis	English	1950	85000000	Fantasy	children's fiction	NaN	NaN	NaN	NaN	Fantasy, children's fiction	Children
164	The Secret Diary of Adrian Mole, Aged 13¾	Sue Townsend	English	1982	20000000	Young adult novel	NaN	NaN	NaN	NaN	NaN	Young adult novel	Young Adult
165	Ronia, the Robber's Daughter	Astrid Lindgren	Swedish	1981	10000000	NaN	NaN	NaN	NaN	NaN	NaN	Children's fiction	Children

```
## Here I am creating a dataframe with the main genres for individual books
# I created a seperate dataframe as books can take multiple genres and this was
the best method to reflect this

genre_class = pd.DataFrame({"genre": ["fantasy", "thriller", "novel",
"autobiographical","biographical","historical",
                  "gothic", "self-help", "romance", "mystery", "philosophical",
"mystery", "science fiction",
                        "non-fiction","fiction"],
                 "count": [df_book["All_Genre"].str.contains("fantasy", case =
False).sum(),
             df_book["All_Genre"].str.contains("thriller", case = False).sum(),
              df_book["All_Genre"].str.contains("novel", case = False).sum(),
          df_book["All_Genre"].str.contains("autobiograph", case = False).sum()
```

```
+ df_book["All_Genre"].str.contains("memoir", case = False).sum(),
             df_book["All_Genre"].str.contains("biograph", case = False).sum(),
           df_book["All_Genre"].str.contains("historical", case = False).sum(),
              df_book["All_Genre"].str.contains("gothic", case = False).sum(),
            df_book["All_Genre"].str.contains("self-help", case = False).sum(),
               df_book["All_Genre"].str.contains("romance", case = False).sum()
+ df_book["All_Genre"].str.contains("romantic", case = False).sum(),
             df_book["All_Genre"].str.contains("mystery", case = False).sum(),
              df_book["All_Genre"].str.contains("romance", case = False).sum()
+ df_book["All_Genre"].str.contains("romantic", case = False).sum(),
            df_book["All_Genre"].str.contains("Philosoph", case = False).sum(),
                 df_book["All_Genre"].str.contains("science fiction", case =
False).sum(),
            df_book["All_Genre"].str.contains("nonfiction", case = False).sum()
+ df_book["All_Genre"].str.contains("non-fiction", case = False).sum(),
             df_book["All_Genre"].str.contains(" fiction", case = False).sum()
+ (df_book["All_Genre"] == "Fiction").sum()]})
```

```
## Here I am plotting the genres in a pie-chart format

# This part allows for spacing between the wedges for more clarity
explode = (0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05, 0.05,
0.05, 0.05, 0.05, 0.05)

# Creating pie-chart with percentages
plt.pie(genre_class["count"], colors=palette15, explode = explode,
        autopct='%1.0f%%', textprops={'size': 'small'})
plt.title( "Pie Chart of Top Genre Classifications for Individual Books")
plt.legend(loc="center    right",    fontsize=10,    title    =    "Genre",
bbox_to_anchor=(1.5,0.5),
           labels = genre_class["genre"])


plt.savefig("../results/genre_pie_chart.png", dpi = 200, bbox_inches = "tight")
```



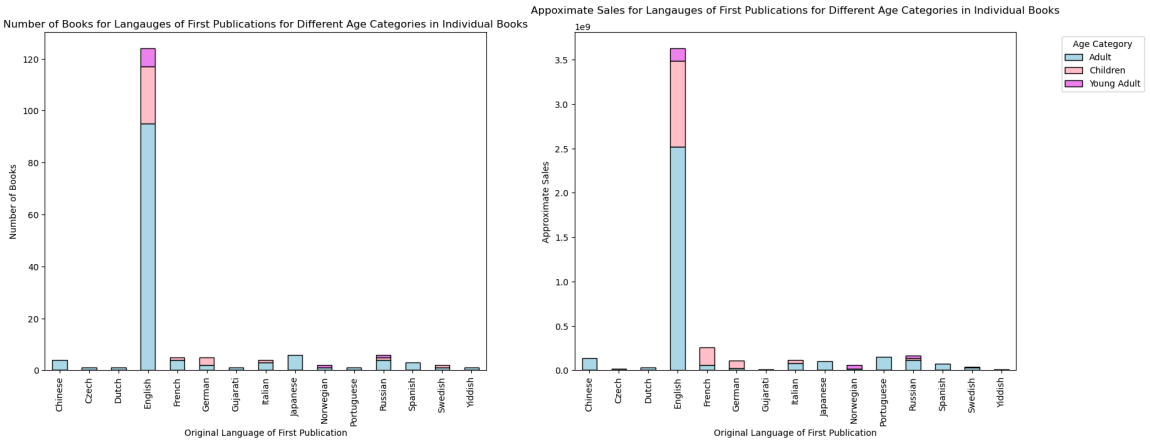
The below plot is an interactive plot that is currently commented out to allow for an easier submission. To see the plot uncomment the lines of code and run the code.

```
## This is an interactive plot for the book genres that cannot be used in the
blog as it is incompatible with hackmd
# However, it is a very insightful plot and is good for investigation and other
uses
# It is a slightly more detailed version of the plot above which is compatible
with interactive plots
# but would be overwhelming on a static plot.

#fig      =      px.pie(genre_class,      values='count',      names='genre',
color="genre",color_discrete_sequence = palette15,
#       title = "Pie Chart of Top Genre Classifications for Individual Books",
#       labels = {"genre": "Book Genre", "count": "Number of Books"})

#fig.show()
```

```
## Here I am making two grouped dataframes for languages and age categories of
plots

# This is a sum of sales dataframe
grouped_age_sales = pd.DataFrame(df_book.groupby([ "Original_Language",
```

```
                                                                "Age_Category"])
['Approximate_Sales'].sum()).unstack()

# This is a count of books dataframe
grouped_age_count = pd.DataFrame(df_book.groupby([ "Original_Language",
                                                                "Age_Category"])
["Book"].count()).unstack()
```

```
## Here I am creating a subplot of two bar plots
# The first is the count of books of original languages split by age category
# The second is the sum of sales of original languages split by age category

fig, axs = plt.subplots(ncols=2, figsize=(20, 7))

grouped_age_count.plot(kind = 'bar', stacked = True, color = ["lightblue",
"pink", "violet"],
                 legend=False, edgecolor = "black", ax = axs[0]).set(
                xlabel = "Original Language of First Publication",
                ylabel = "Number of Books",
               title = "Number of Books for Langauges of First Publications for
Different Age Categories in Individual Books"
                )

grouped_age_sales.plot(kind = 'bar', stacked = True, color = ["lightblue",
"pink", "violet"],
                 legend=False, edgecolor = "black", ax = axs[1]).set(
                xlabel = "Original Language of First Publication",
                ylabel = "Approximate Sales"
                )
plt.legend(["Adult",   "Children",   "Young   Adult"],   loc='upper   right',
bbox_to_anchor=(1.3, 1), title = "Age Category")

plt.title("Appoximate Sales for Langauges of First Publications for Different
Age Categories in Individual Books", y =1.04)


plt.savefig("../results/book_language_age.png", bbox_inches = "tight")
```




```
## Here I am making a subplot of two scatter graphs
# They are of years of first published against approximate sales
# The colours map to the original language
# The second plot is a subsection of the first plot

fig, axs = plt.subplots(ncols=2, figsize=(15, 7))
sns.scatterplot(data=df_book, y="Approximate_Sales", x="First_Published",
                hue="Original_Language", palette = palette16, ax = axs[0],
legend=False).set(
                 xlabel = "Year of First Publication", ylabel = "Approximate
Sales"
                )

sns.scatterplot(data=df_book, y="Approximate_Sales", x="First_Published",
            hue="Original_Language", palette = palette16, ax = axs[1]).set(
                xlabel = "Year of First Publication",
                yticklabels=[],
                ylabel = None,
                xlim = [1900, 2025]
                )

plt.legend(loc='upper right', bbox_to_anchor=(1.3, 1), title = "Original
Language")

fig.suptitle("Appoximate Sales across Years of Publications for Individual
Books")

plt.savefig("../results/book_language_scatter.png", bbox_inches = "tight")
```

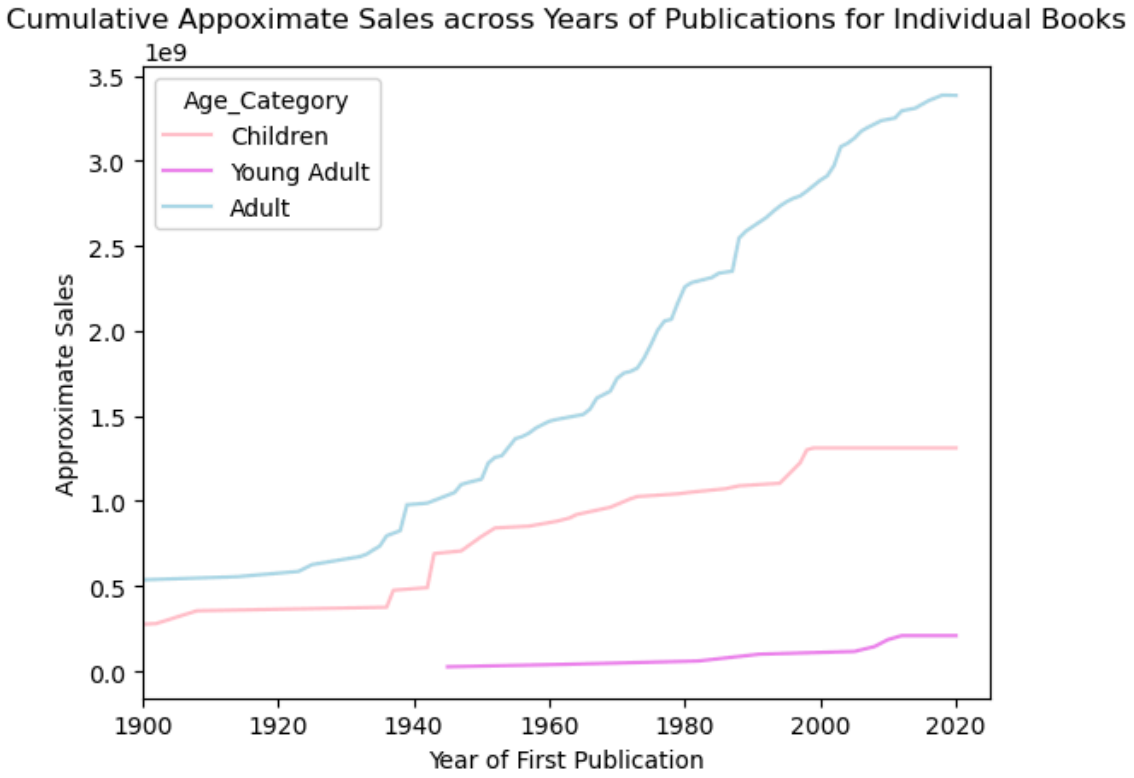
The below plot is an interactive plot that is currently commented out to allow for an easier submission. To see the plot uncomment the lines of code and run the code.

```
## This is an interactive plot of above that cannot be used in the blog as it
is incompatible with hackmd
# However, it is a very insightful plot and is good for investigation and other
uses
# It is a slightly more detailed version of the plot above which is compatible
with interactive plots
# but would be overwhelming on a static plot.

#fig2 = px.scatter(df_book, x="First_Published", y ="Approximate_Sales",
#             color="Original_Language", color_discrete_map=palette16,
#          title = "Appoximate Sales across Years of Publications for Individual
Books",
#             hover_data=["Book", "Author"],
#             labels={ "First_Published": "Year of First Publication",
#                     "Original_Language": "Original Language of Publication",
#                     "Approximate_Sales": "Approximate Sales"
#                 })

#fig2.show()
```

```
## Here I am making a cummulative sum dataframe of the different age categories
across years

children = df_book[df_book["Age_Category"] == "Children"][["First_Published",
"Approximate_Sales"]
                                                    ].groupby("First_Published").sum().cumsum().reset_index()
```

```
children["Age_Category"] = "Children"

young_adult    =    df_book[df_book["Age_Category"]    ==    "Young    Adult"]
[["First_Published", "Approximate_Sales"]
                                          ].groupby("First_Published").sum().cumsum().reset_index
young_adult["Age_Category"] = "Young Adult"

adult   =   df_book[df_book["Age_Category"]   ==   "Adult"][["First_Published",
"Approximate_Sales"]
                                    ].groupby("First_Published").sum().cumsum().reset_index()
adult["Age_Category"] = "Adult"

# Here I am adding the maximum amounts as the last datapoint to make sure the
sums finish at the same point
final_entries = pd.DataFrame({"First_Published": [2020, 2020, 2020],
                                    "Approximate_Sales": [1312000000, 209000000,
3383900000],
                                      "Age_Category":["Children", "Young Adult",
"Adult"]})

age_range_sales = pd.concat([children, young_adult, adult, final_entries])
```

```
## Here I am plotting the cummualtive sums of sales of the three age categories
# This is a line graph as it is best suited to a time-series analyis

sns.lineplot(age_range_sales, x="First_Published", y ="Approximate_Sales",
            hue="Age_Category", palette = ["pink", "violet", "lightblue"]).set(
                xlabel = "Year of First Publication",
                ylabel = "Approximate Sales",
                xlim = [1900, 2025]
                )

plt.title(  "Cumulative  Appoximate  Sales  across  Years  of  Publications  for
Individual Books", y =1.04)

plt.savefig("../results/cumulative_sales_plot.png", bbox_inches = "tight")
```



Within this section, I will be creating plots for the series dataframe.

```
## Here I am loading in the book series dataframe and checking the format
df_series = pd.read_csv('../data/top_series.csv')
df_series = df_series.drop(df_series.columns[0], axis =1)
df_series
```

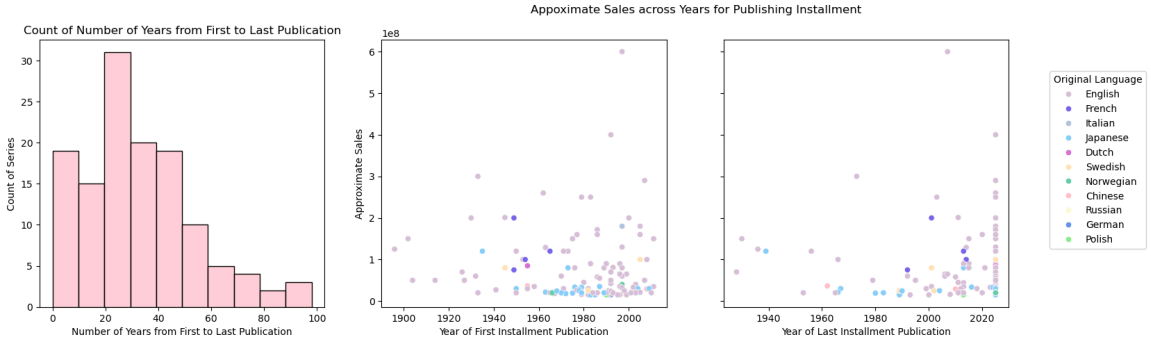
	Book_Series	Author	Original_Language	No_of_Books	Year_of_Publication	Approximation_Sales	Age_Category	Genre_1	Genre_2	First_Installment_Published	Last_Installment_Published	No_of_Main_Books	No_of_Extrabooks	Extra_Books	All_Years	Number_of_Years_Published
0	Harry Potter	J. K. Rowling	English	7 + 3 companion books + 4 scripts	1997–2007	600000000	NaN	NaN	NaN	1997	2007	7.0	7.0	True	NaN	10
1	Goosebumps	R. L. Stine	English	62 + spin	1992–2025	400000000	NaN	NaN	NaN	1992	2025	62.0	NaN	True	NaN	33

Book_Series	Author	Original_Language	Years_of_Installments	Num_of_Publications	Approximation_Sales	Genre1	Genre2	First_Installment_Published	Last_Installment_Published	No_of_Main_Books	No_of_Extrasbetween	Extra_Books	All_Years_Published	Number_of_Publication
			off series											
2	Perry Mason	Erle Stanley Gardner	English	82 + 4 short stories	1933–1973	300000000	NaN	NaN	1933	1973	82.0	4.0	True	NaN	40
3	Diary of a Wimpy Kid	Jeff Kinney	English	19 + 5 spin-offs	2007–2025	290000000	NaN	NaN	2007	2025	19.0	5.0	True	NaN	18
4	Berenstain Bears	Stan and Jan Berenstain	English	428	1962–2025	260000000	NaN	NaN	1962	2025	428.0	NaN	False	NaN	63
…	…	…	…	…	…	…	…	…	…	…	…	…	…	…	…
123	Horrible Histories	Terry Deary	English	24	1993–2025	20000000	NaN	NaN	1993	2025	24.0	NaN	False	NaN	32
124	Rainbow Magic	Daisy Meadows	English	80	2003–2025	20000000	NaN	NaN	2003	2025	80.0	NaN	False	NaN	22
125	Morgan Kane	Louis Masterson	Norwegian	90	1966–2025	20000000	NaN	NaN	1966	2025	90.0	NaN	False	NaN	59
126	The Southern Vampire Mysteries	Charlaine Harris	English	13	2001–2013	20000000	NaN	NaN	2001	2013	13.0	NaN	False	NaN	12

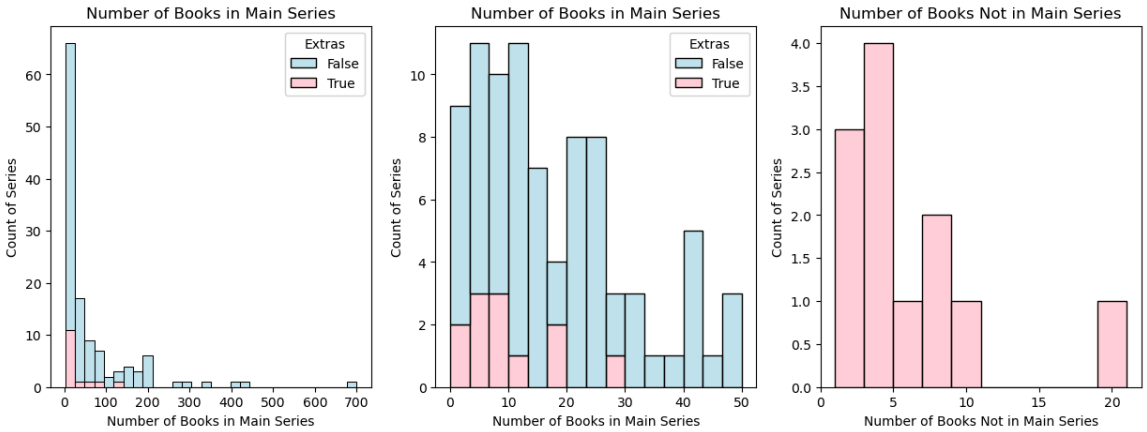
	Book_Series	Author	Original_Language	Number_of_Years_Publication	Year_of_Last_Publication	Approximate_Sales	Genre	Genre_1	First_Installment_Published	Last_Installment_Published	No_Of_Main_Books	No_of_Extrabetrasbetras	Extra_Years	All_Genres	Number_Publication
127	鬼平犯科帳 (Onihei Hankachō)	Shōtarō Ikenami	Japanese	24	1990	24400000	jidaigeki	NaN	1968	1990	24.0	NaN	False	jidaigeki	22

```
## Here I am creating a subplot with one histogram and two scatter graphs
# They use the same colour schemes that are used within the individual book visualisations

fig, axs = plt.subplots(ncols=3, figsize=(18, 5))

sns.histplot(df_series, x = "Number_of_Years_Publication", color ="pink" , ax = axs[0]).set(
                xlabel = "Number of Years from First to Last Publication",
                ylabel = "Count of Series",
              title = "Count of Number of Years from First to Last Publication")

sns.scatterplot(data=df_series,                          y="Approximate_Sales", x="First_Installment_Published",
                    hue="Original_Language", palette = palette16, ax=axs[1], legend=False).set(
                xlabel = "Year of First Installment Publication",
                ylabel = "Approximate Sales",
                )

sns.scatterplot(data=df_series,                          y="Approximate_Sales", x="Last_Installment_Published",
            hue="Original_Language", palette = palette16, ax = axs[2]).set(
                yticklabels=[],
                ylabel = None,
                xlabel = "Year of Last Installment Publication"
                )

plt.legend(loc='upper right', bbox_to_anchor=(1.5, 0.9), title = "Original Language")

plt.suptitle( "Appoximate Sales across Years for Publishing Installment", x =
```

```
0.65)

plt.savefig("../results/first_to_last_installment_plot.png",    bbox_inches    =
"tight")
```




```
## Here I am making a subplot of three histograms
# They are all related to number of book fields
# The last two use the Boolean field for extra books as a colour category

fig, axs = plt.subplots(ncols=3, figsize=(15, 5))

sns.histplot(df_series, x = "No_of_Main_Books", stat = "count",
             bins = 30, hue='Extras', multiple='stack', palette=['lightblue',
'pink'],
             ax=axs[0]).set(
                xlabel = "Number of Books in Main Series",
                ylabel = "Count of Series",
                title = "Number of Books in Main Series")

sns.histplot(df_series, x = "No_of_Main_Books", stat = "count",
              bins = 15, binrange= [0,50], hue='Extras', multiple='stack',
palette=['lightblue', 'pink'],
             ax=axs[1]).set(
                xlabel = "Number of Books in Main Series",
                ylabel = "Count of Series",
                title = "Number of Books in Main Series")

sns.histplot(df_series, x = "No_of_Extras", color ="pink" , bins = 10, ax =
axs[2]).set(
                xlabel = "Number of Books Not in Main Series",
                ylabel = "Count of Series",
                title = "Number of Books Not in Main Series")

plt.savefig("../results/number_books_series.png", bbox_inches = "tight")
```

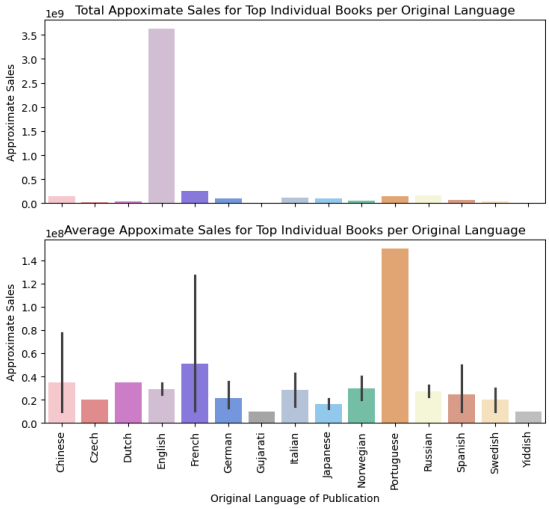



```
## Here I am creating a summary table on number of books in the series

table_series                    =                    df_series[["No_of_Main_Books",
"No_of_Extras"]].describe().round(1)
table_series.columns = ["Number of Main Books", "Number of Extra Books"]
table_series = table_series.transpose()
table_series.columns = ["Count of Non Null Values", "Mean", "Standard Deviation",
"Minimum", "Lower Quantile (25%)",
                        "Median (50%)", "Upper Quantile (75%)", "Maximum"]

table_series.to_csv('../results/table_series.csv')
```

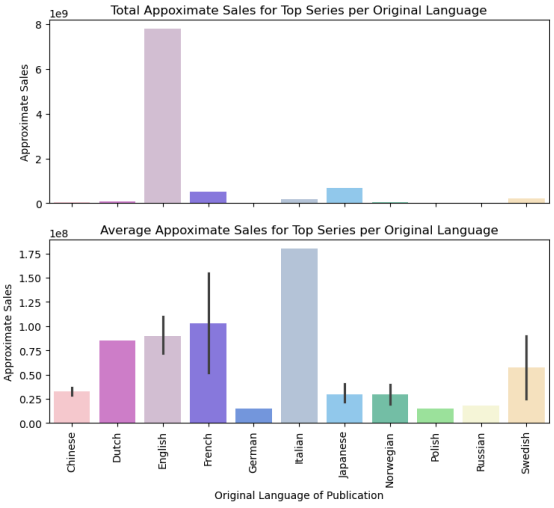
This last section combines a plot for the series and individual books.

```
## Here I am making a subplots of 4 bar graphs
# Two are from the individual book data
# The other two are from the book series data
# They are made in a similar format to eachother
# I create two grouped tables to get a sum plot and the main dataframes to get
plots of the mean sales

fig, axs = plt.subplots(nrows=2, ncols=2, figsize=(19, 7))

grouped_sales_language = df_book.groupby("Original_Language").sum()
g1   =   sns.barplot(grouped_sales_language,   x   =   "Original_Language",   y   =
"Approximate_Sales", hue = "Original_Language",
         palette = palette16, legend = False, errorbar=None, ax = axs[0, 0],
                                                                  order      =
df_book.sort_values("Original_Language").Original_Language).set(
                xticklabels=[],
                xlabel = None,
                ylabel = "Approximate Sales",
                  title = "Total Appoximate Sales for Top Individual Books per
```

```
Original Language")


g2 = sns.barplot(df_book, x = "Original_Language", y = "Approximate_Sales", hue
= "Original_Language",
                            palette=  palette16,  legend  =  False,  order  =
df_book.sort_values("Original_Language").Original_Language,
             ax = axs[1, 0] )

g2.tick_params(axis = 'x', rotation=90)

g2.set( xlabel = "Original Language of Publication",
                ylabel = "Approximate Sales",
                 title = "Average Appoximate Sales for Top Individual Books per
Original Language"
                )

grouped_sales_language = df_series.groupby("Original_Language").sum()
g3  =  sns.barplot(grouped_sales_language,  x  =  "Original_Language",  y  =
"Approximate_Sales", hue = "Original_Language",
         palette = palette16, legend = False, errorbar=None, ax = axs[0, 1],
                                                                  order     =
df_series.sort_values("Original_Language").Original_Language).set(
                xticklabels=[],
                xlabel = None,
                ylabel = "Approximate Sales",
                  title = "Total Appoximate Sales for Top Series per Original
Language")

g4 = sns.barplot(df_series, x = "Original_Language", y = "Approximate_Sales",
hue = "Original_Language",
                            palette=  palette16,  legend  =  False,  order  =
df_series.sort_values("Original_Language").Original_Language,
             ax = axs[1, 1] )

g4.tick_params(axis = 'x', rotation=90)

g4.set(xlabel = "Original Language of Publication",
                ylabel = "Approximate Sales",
                 title = "Average Appoximate Sales for Top Series per Original
Language")
plt.savefig("../results/sales_per_language.png", bbox_inches = "tight")
```


Total Appoximate Sales for Top Individual Books per Original Language
1e9
Approximate Sales
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5

Average Appoximate Sales for Top Individual Books per Original Language
1e8
Approximate Sales
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Chinese
Czech
Dutch
English
French
German
Gujarati
Italian
Japanese
Norwegian
Portuguese
Russian
Spanish
Swedish
Yiddish
Original Language of Publication


Total Appoximate Sales for Top Series per Original Language
1e9
Approximate Sales
0
2
4
6
8

Average Appoximate Sales for Top Series per Original Language
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Approximate Sales
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0.25
0.50
0.75
1.00
1.25
1.50
1.75
Chinese
Dutch
English
French
German
Italian
Japanese
Norwegian
Polish
Russian
Swedish
Original Language of Publication